

Supplementary Material E: Facilitator Selection, Training, and Supervision

Note. This document was compiled from the training program records and supervision logs (compiled August 2026). It is anonymized for peer review. Certificates of completion for the three external curricula (four facilitators $\times$ three courses) contain facilitator-identifying information and are therefore not included in this anonymized document; records of completion are available from the corresponding author upon reasonable request.

E1. Selection

The four facilitators were recent graduates of Electronic Engineering (mean post-graduation experience: 14 months, $SD = 4.2$), selected on the basis of demonstrated functional proficiency in Python, ChatGPT, and Google Colab, verified through a structured skills assessment administered by the principal investigators prior to training.

E2. Pre-intervention training program (41 hours)

All four facilitators completed a structured 41-hour training program delivered across the four weeks immediately preceding the study (Table E1).

Table E1: Components of the 41-hour facilitator training program.

Component	Provider	Hours	Dimension alignment
<i>AI for Everyone</i>	DeepLearning.AI / Coursera	7	D1, D3
<i>AI Agents in Python</i>	DeepLearning.AI / Coursera	8	D2; technological mediation
<i>AI Fluency: Framework & Foundations</i> (4D framework)	Anthropic Academy	3	D3, D4
<i>AI Fluency for Educators</i>	Anthropic Academy	1	D3, D4
<i>Building with the Claude API</i>	Anthropic Academy	2	D2
<i>AI Safety Fundamentals</i> (Constitutional AI, value alignment, systemic bias)	Anthropic Academy	4	D3
Internal workshop: five-phase session protocol	Research team	4	All components
Internal workshop: Structured Academic Controversy facilitation	Research team	3	D3
Internal workshop: portfolio-based assessment design	Research team	3	D4; connectivist assessment
Total		41	

E3. Competency verification

At the conclusion of training, facilitators were assessed with the same 15-indicator checklist used for implementation fidelity monitoring (Supplementary Material D). All four demonstrated comparable competency profiles (mean pre-intervention competency score: 84%; range: 81%–88%).

E4. Rotation and supervision during implementation

A systematic rotation design ensured that each facilitator delivered sessions across multiple sections, preventing confounding of individual facilitator characteristics with treatment effects within the experimental condition. Weekly supervision meetings with the principal investigators (minimum 45 minutes) reviewed session logs and recalibrated facilitator practices as needed throughout the intervention period. Each facilitator maintained a per-session log.

E5. Control-group instructor

The control sections were delivered by a separate instructor with comparable qualifications who received the same standardized content guide to ensure curricular equivalence, but who was not trained in, or exposed to, any component of the connectivist framework. The control-group instructor did not attend the training program described in Table E1 and did not participate in the weekly supervision meetings, which were specific to the experimental condition. This difference in personnel and supervision across conditions is acknowledged as a limitation in the main text.